

| PhD thesis in Signal Processing

Title *Bayesian Estimate of Galaxy Clusters for Improved Cosmological Parameter Inference*
Location *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRISTAL, SigMA team), Villeneuve-d'Ascq, France*
Supervisors [Jenny Sorce](#) and [Pierre-Antoine Thouvenin](#)
Funding ANR Chaire WILL UNIVERSITWINS Project hold by [Jenny Sorce](#) and [Klaus Dolag](#)
Keywords Inverse problem solving, Bayesian hierarchical model, generative models, MCMC algorithms.

| Research internships

Apr. 2025 - **Pre-PhD Research internship** *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRISTAL, SigMA team), Villeneuve-d'Ascq, France*
Sep. 2025
Title Bayesian Estimate of Galaxy Clusters for Improved Cosmological Parameter Inference
Supervisors [Jenny Sorce](#) and [Pierre-Antoine Thouvenin](#)
Funding ANR Chaire IA SHERLOCK hold by [Pierre Chainais](#)
Keywords Inverse problem solving, Bayesian inference, X-ray spectra modeling.

Apr. 2024 - **Research internship**, *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRISTAL, SigMA team), Villeneuve-d'Ascq, France*
Jul. 2024
Title Galaxy cluster mass estimate using radial velocities
Supervisors [Jenny Sorce](#) and [Pierre-Antoine Thouvenin](#)
Funding ANR Chaire IA SHERLOCK hold by [Pierre Chainais](#)
Keywords Inverse problem solving, physic based inference, Block Coordinate Forward Backward algorithm.

| Education

2023 - 2025 **Master Degree in Data Science** (rank: 2/19), *Université de Lille, Lille, France*
Research-oriented Master entirely taught in english.

2020 - 2023 **Bachelor Degree in Applied Mathematics** (rank: 2/28), *Université de Lille, Lille, France*
Multidisciplinary bachelor focused on mathematics, economics and computer science.

| Research projects

Title Mapping the distance between the simulated and the observed cosmic webs
Year Master 2
Supervisors [Jenny Sorce](#) and [Jérémy Boulanger](#)
Keywords Constrained simulations, separation metrics to define dissimilarities: likelihood, Mahalanobis and Sinkhorn distance.

Title Galaxy cluster mass estimate using radial velocities
Year Master 1
Supervisors [Jenny Sorce](#) and [Pierre-Antoine Thouvenin](#)
Keywords Inverse problem solving, pattern recognition, Gaussian Processes.

Title Introduction to Topological Data Analysis (TDA)
Year 3rd y. Bachelor
Supervisors [Camille Sabbah](#)
Keywords Basic notions of topology, persistent homology, mapper algorithm, Betti number.

| Languages and programming Skills

Languages French (Native), English (C1 level), German (B1 level)
Programming Python, Git, R, Git, LaTeX, SQL, HTML
Softwares: PowerBi, Excel